

4. Bias-Variance Trade-off

Problem 4: Predicting Energy Industry Returns with Increasing Model Complexity

Research Anchor: Geman, Bienenstock and Doursat (1992), “Neural Networks and the Bias/Variance Dilemma”

Dataset: Kenneth R. French Data Library — Fama/French 3 Factors and 10 Industry Portfolios

This problem examines the bias-variance trade-off using a financial return prediction exercise. The objective is to understand whether increasing model complexity improves prediction performance or leads to overfitting.

Using the Energy industry excess return as the target variable and market excess return as the predictor, polynomial regression models of increasing degrees are estimated and compared using training RMSE and test RMSE. This helps show the difference between fitting past data well and making reliable predictions on unseen data.

Situation	Meaning
High Bias	Model is too simple. It underfits.
High Variance	Model is too complex. It overfits.
Good balance	Model performs reasonably well on both training and test data.

Table 1: Bias-Variance Intuition

Findings

Our best test RMSE is:

- Polynomial Degree = 7
- Test RMSE = 5.839960

But compare it with Degree 1:

- Degree 1 Test RMSE = 5.850828
- Degree 7 Test RMSE = 5.839960
- Difference = 0.010868

That difference is extremely small. So even though Degree 7 is technically the best in this table, the **improvement over simple linear regression is almost negligible.**

Polynomial Degree	Training RMSE	Test RMSE
1	3.672584	5.850828
2	3.672556	5.850901
3	3.671772	5.852974
4	3.661544	5.842743
5	3.659139	5.853509
6	3.659022	5.855362
7	3.652892	5.839960
8	3.648011	5.858788
9	5.094968	6.658766
10	5.030356	6.547669

Table 2: Bias-Variance Trade off Results

Degree range	What happens?
1 to 4	Test RMSE stays almost the same
5 to 8	Test RMSE still stays around the same level
9 to 10	Test RMSE jumps upward

Table 3: Interpretation of Model Complexity in terms of Polynomial Degrees

This tells us that **increasing model complexity does not meaningfully improve prediction** of Energy excess returns from market excess returns.

Low-degree models: Degree 1 to 2: These are simple models. They have:

- lower complexity
- higher bias
- lower variance

But in our case, Degree 1 already performs quite well relative to the more complex models. So, the simple model is not badly underfitting.

Medium-degree models: Degree 3 to 8: These allow more curvature. Training RMSE falls slightly:

- Degree 1 Training RMSE = 3.6726
- Degree 8 Training RMSE = 3.6480

So, the model fits the training data slightly better. But test RMSE barely improves:

- Degree 1 Test RMSE = 5.8508
- Degree 8 Test RMSE = 5.8588

This means, the extra flexibility is mostly fitting small patterns in the training data, but those patterns do not generalise strongly to the test data.

High-degree models: Degree 9 to 10: Here, both training RMSE and test RMSE jump.

- Degree 9 Test RMSE = 6.6588
- Degree 10 Test RMSE = 6.5477

This tells us that very high-degree polynomial models are unstable and perform worse. Very complex models can become unreliable and perform badly on unseen data.

Which model should we choose?

Technically, the lowest test RMSE is Degree 7. But practically, I would choose **Degree 1**, because:

- Degree 1 is much simpler
- Degree 1 is easier to interpret
- Degree 1 performs almost as well as Degree 7
- Degree 7 gives only a tiny improvement

This is an important modelling principle: Do not choose a more complex model unless the improvement is meaningful. The simple linear model is preferable because it is interpretable and performs almost as well as the more complex polynomial models.

Interpretation

This exercise examines the bias-variance trade-off by predicting Energy industry excess returns using market excess returns with increasing polynomial model complexity. Polynomial degrees from 1 to 10 were tested, and performance was evaluated using training RMSE and test RMSE.

The results show that increasing model complexity reduces training error only slightly from Degree 1 to Degree 8. However, the test RMSE remains almost unchanged across these models. The simple Degree 1 model has a test RMSE of 5.8508, while the best-performing model, Degree 7, has a test RMSE of 5.8400. This improvement is very small and does not justify the additional complexity of the higher-degree model.

For Degrees 9 and 10, test RMSE increases sharply, suggesting that very high-complexity models perform worse on unseen data. This illustrates the risk of overfitting and numerical instability when models become unnecessarily complex.

Overall, the results show that more complexity does not automatically lead to better prediction. In this case, a simple linear model is preferable because it is easier to interpret and performs almost as well as the more complex polynomial models. This exercise helps explain why regularization methods such as Ridge, LASSO, and Elastic Net are useful: they control model complexity and help reduce overfitting.

Learnings / Errors

There was one technical issue I fixed before writing the final interpretation. From Degree 1 to 4, the test error is almost flat. Degree 4 is slightly better, but only by a tiny amount. This means increasing the model complexity from Degree 1 to Degree 4 does not meaningfully improve prediction. Then from Degree 5 onwards, the test error becomes much worse. This means the more complex polynomial models are performing worse on unseen data. That is the classic warning sign of **overfitting**.

One important issue: In theory, as polynomial degree increases, the **training RMSE should not increase**. A more flexible model should be able to fit the training data at least as well as a simpler model. But in our result, training RMSE jumps. This is not the usual theoretical behaviour. The likely reason is **numerical instability**. Why? Because polynomial terms become very large. For example, if $\text{Mkt-RF} = 15$, then:

$$15^5 = 759,375$$

So, the regression struggles with huge numbers. This can distort the result. Therefore, before polynomial regression, we should **standardise the input variable**.

After scaling, the training RMSE should generally go down or stay almost the same as degree increases. The test RMSE may behave differently:

- Training RMSE goes down = model fits training data better
- Test RMSE goes up = model is overfitting

That is the **Bias-Variance trade-off**.

Training error tells us how well the model fits the data it has already seen. Test error tells us how well the model performs on unseen data. A model that performs well on training data but badly on test data is probably Overfitting.

References and Bibliography

Geman, S., Bienenstock, E. and Doursat, R. (1992) 'Neural Networks and the Bias/Variance Dilemma', *Neural Computation*, 4(1), pp. 1–58. DOI: 10.1162/neco.1992.4.1.1.

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Appendix

A.4 Problem 4: Bias-Variance Trade-off in Energy Return Prediction Using Polynomial Regression

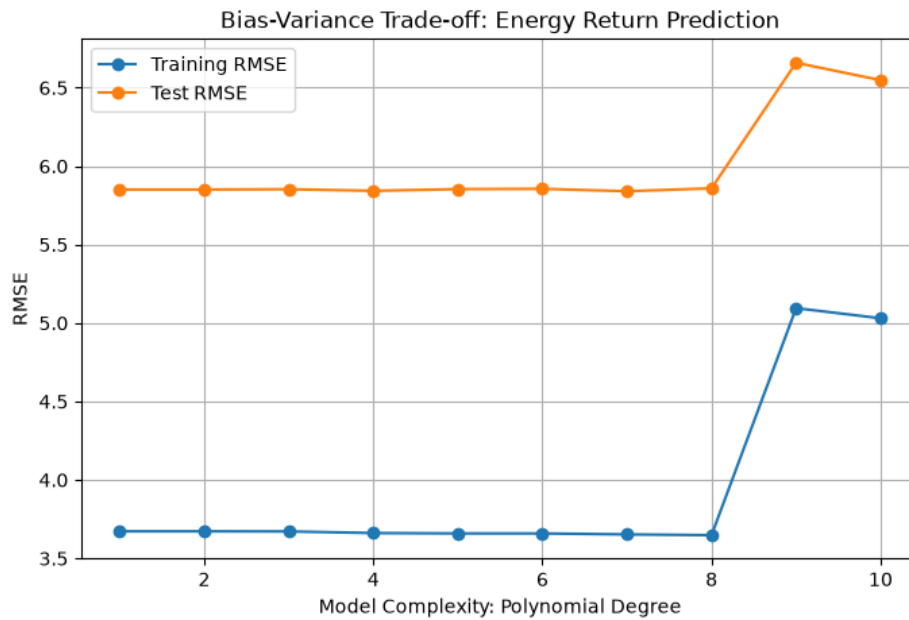


Figure 1: Training and Test RMSE Across Increasing Polynomial Model Complexity.